Intuition → diff bits → XOR = 1

Ques: Single Number III

arr = { 3, 4, 3, 2, 4, 7, 8, 9, 2, 8 }

b1 = 0 ^ ~~3~~ ^ ~~3~~ ^ ~~2~~ ^ 7 ^ ~~2~~ = 7 (non-zero)

b2 = 0 ^ ~~4~~ ^ ~~4~~ ^ ~~8~~ ^ 9 ^ ~~8~~ = 9 (zero)

Method-2 XOR

7 ^ 9

```
  0111
^ 1001
 ------
  1110   = 14
```

↓

mask = 0010 → is diff in 2 single numbers

```
  1001
& 0010
 ------
  0000
```

```
  101100   n
& 101011   n-1
 ------
  101000
 ------
^ 101100   n
 ------
  000100   mask
```

↓

rmsb

[n & (n-1)] ^ n

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Ques: XOR of numbers in a given range

$1 \wedge 2 \wedge 3 \wedge 4 \wedge 5 \wedge 6 \wedge 7 \ldots \wedge n = ?$

Method-1: loop TC = $O(n)$

Method-2: TC = $O(1)$

$n \% 4$	$1 \ldots \wedge n$
1	1
2	$n+1$
3	0
0	n

$n \wedge (n+1) = 1$

$n \to$ even, $(n+1) \to$ odd

$n \wedge 1 = n+1$

$n \to$ even

1	=	1
$1 \wedge 2$	=	3
$1 \wedge 2 \wedge 3$	=	0
$1 \wedge 2 \wedge 3 \wedge 4$	=	4
$1 \wedge 2 \wedge 3 \wedge 4 \wedge 5$	=	1
$1 \ldots 6$	=	7
$1 \ldots 7$	=	0
$1 \ldots 8$	=	8
$1 \ldots 9$	=	1
$1 \ldots 10$	=	11
$1 \ldots 11$	=	0
$1 \ldots 12$	=	12

even $\uparrow$

$n \wedge 1$

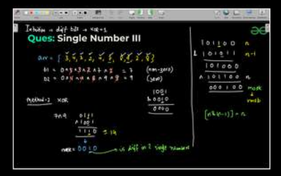
$= n+1$

$1 \ldots 13 = 1$

$1 \ldots 14 = 15$

$1 \ldots 15 = 0$

$1 \ldots 16 = 16$

$n \wedge 0 = n$

$n \wedge n = 0$